



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 -Electricity Markets and Energy System Planning, Specialisation Course

**Stochastic market clearing (two-stage
stochastic programming)**

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Professor

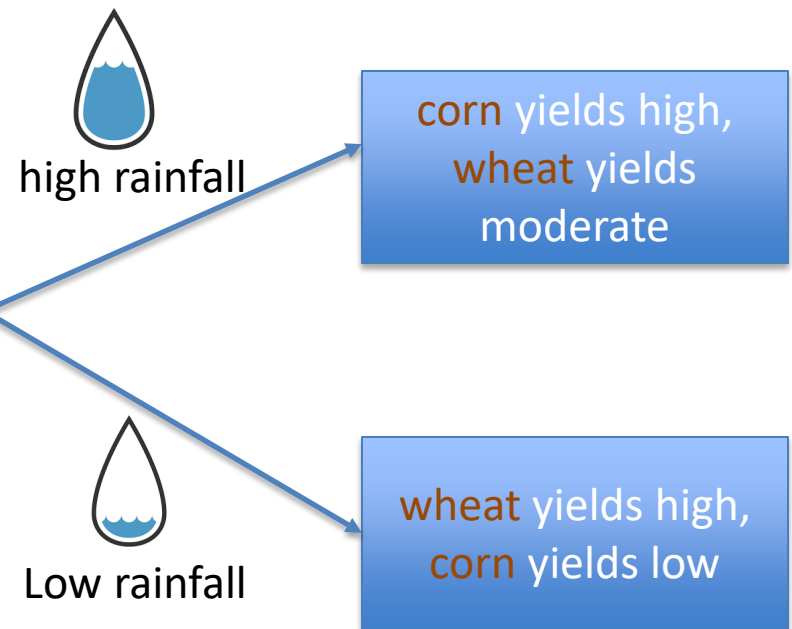
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Overview

- Introduction to examples of a two-stage problem under uncertainty
- Clearing the Day-Ahead Market Under Uncertainty (Background)
- Stochastic Programming Problems with Recourse (two-stage)
- Mathematical formulation: Stochastic market-clearing

Example: Farmer's Crop Planning

- A farmer must decide how much land to allocate to **wheat** and **corn**.
- The weather (**rainfall**) is uncertain, which affects yields and profits.



Uncertainty



If ... → then ...

Two-stage problem

Stage 1- Before planting (Here-and-now decision)	Stage 2 – After weather is realized (Recourse decision):
• The farmer decides how many hectares to allocate to wheat and corn. • Decision must be made before <u>knowing the weather</u>.	The farmer can adapt by: • Buying extra feed/grain if crops fail (at market price). • Selling surplus crops if yields are high.

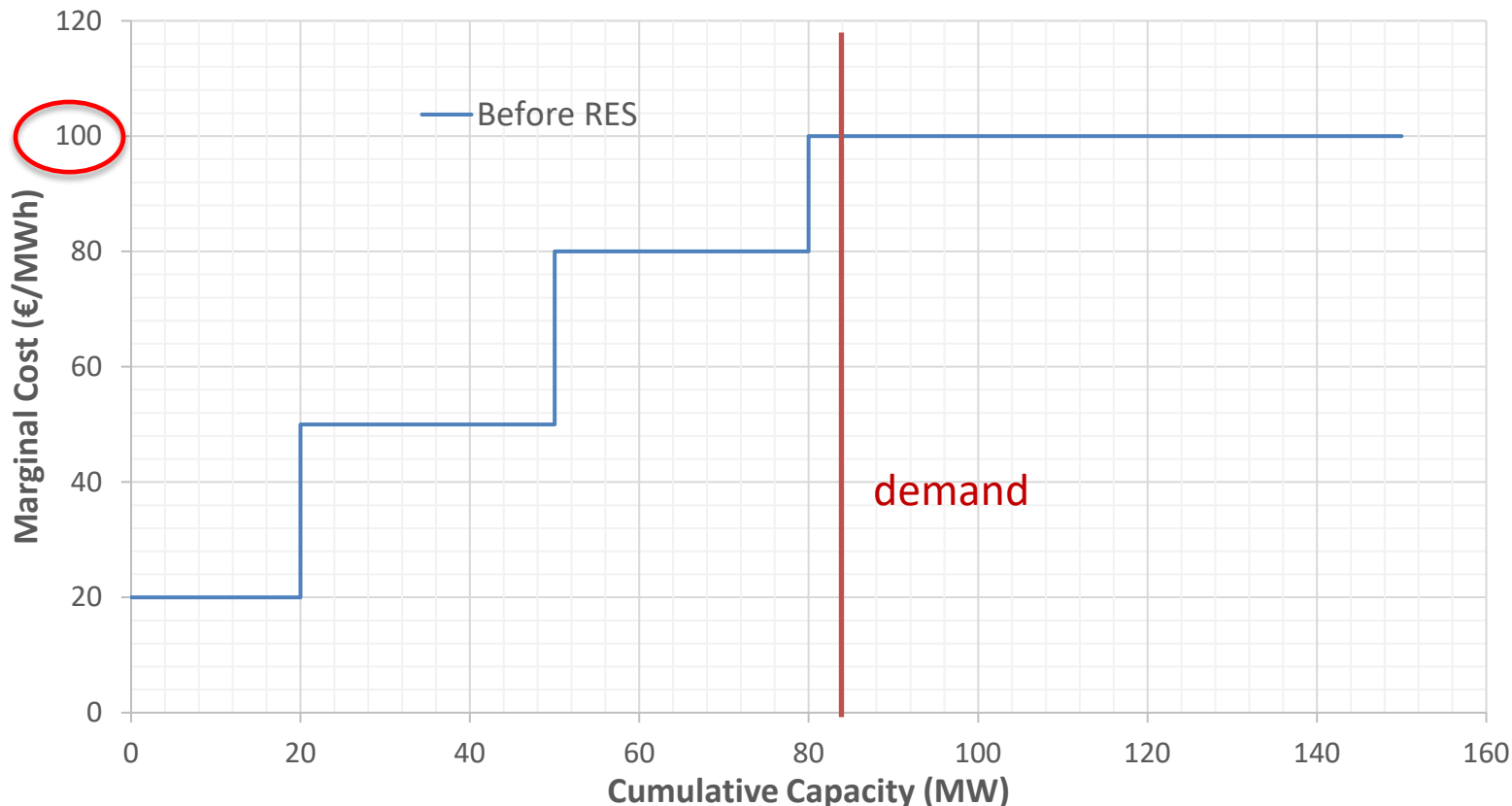


- **First-stage:** Land allocation is fixed before uncertainty is revealed.
- **Second-stage:** Actions (buying/selling crops) depend on actual rainfall.

Back to Electricity Market:

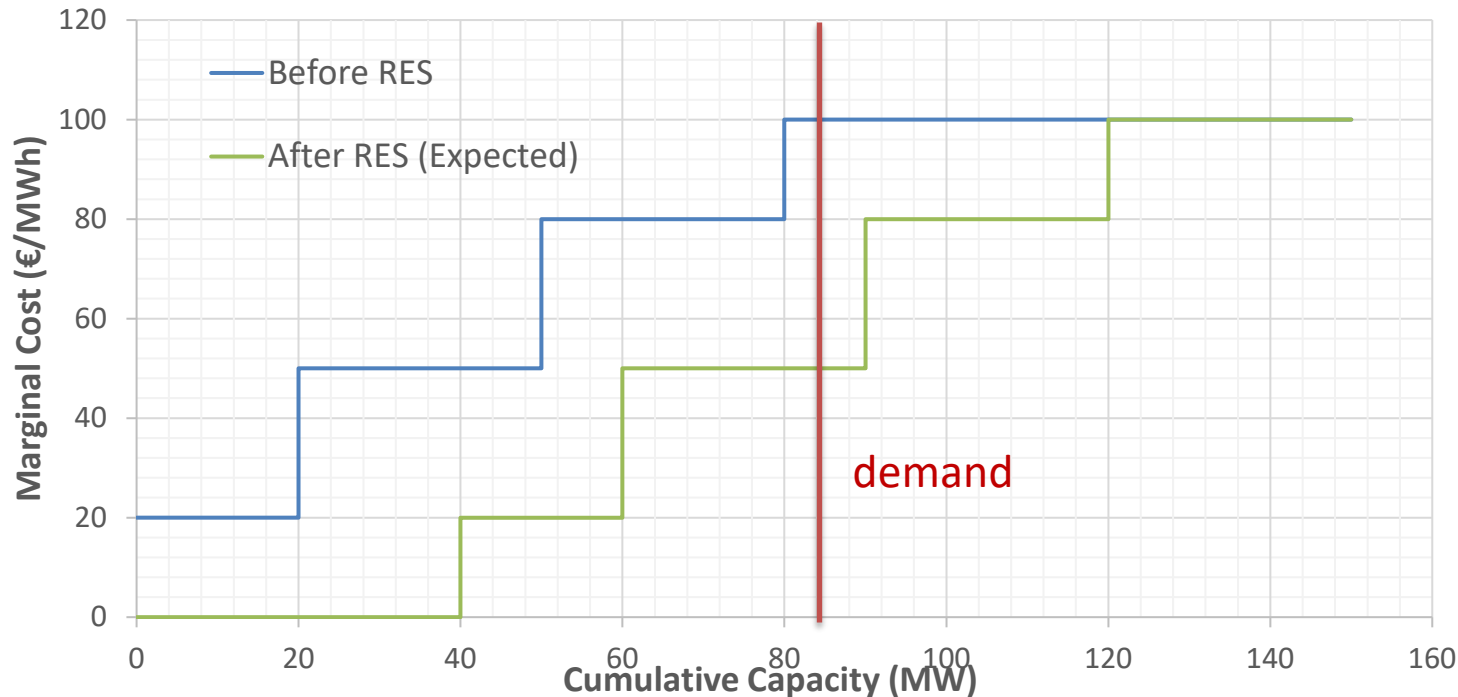
Impact of RES on the Merit Order

- Electricity producers are ranked by marginal cost (€/MWh).
- Conventional order: Nuclear / Hydro → Coal → Gas → Oil → Peaking units.
- Market clearing price set by last unit dispatched (marginal plant).



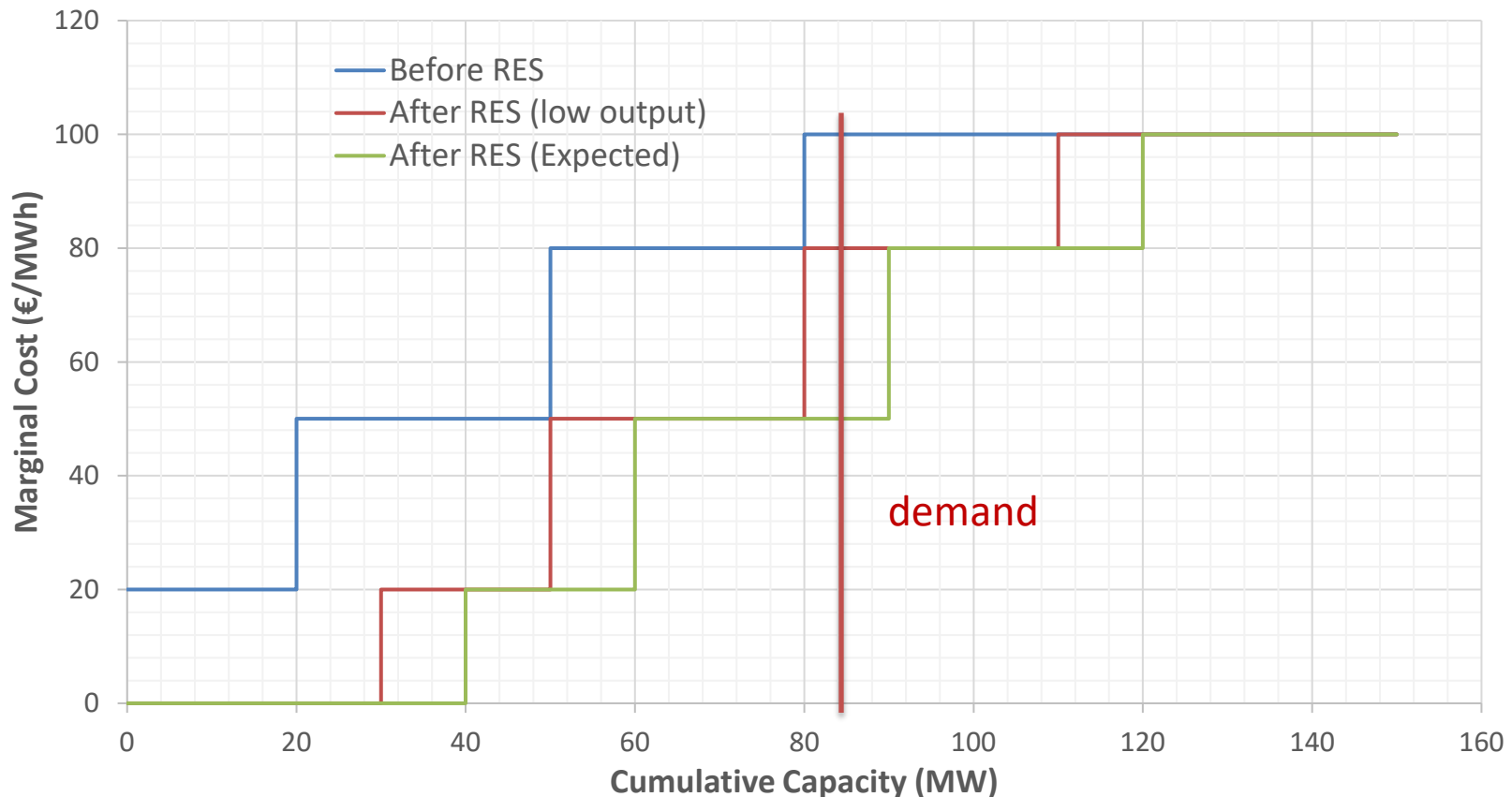
Impact of RES on the Merit Order

- RES enter at near-zero marginal cost → shift merit order to the right.
- Effect:
 - Lower market-clearing price (merit-order effect).
 - Fossil fuel plants displaced (reduced operating hours).
 - Greater price volatility due to weather dependency.



Impact of RES on the Merit Order

- RES integration reduces wholesale electricity prices
- But increases uncertainty and requires flexible backup



Connection Between Farmer's Crop Planning and Energy Systems

- Farmers committing land = power producers committing generation in the day-ahead market.
- Rainfall outcome = uncertain renewable generation (wind/solar inflow).
- Buying/selling crops = balancing market adjustments.

Uncertainty as a part of market!

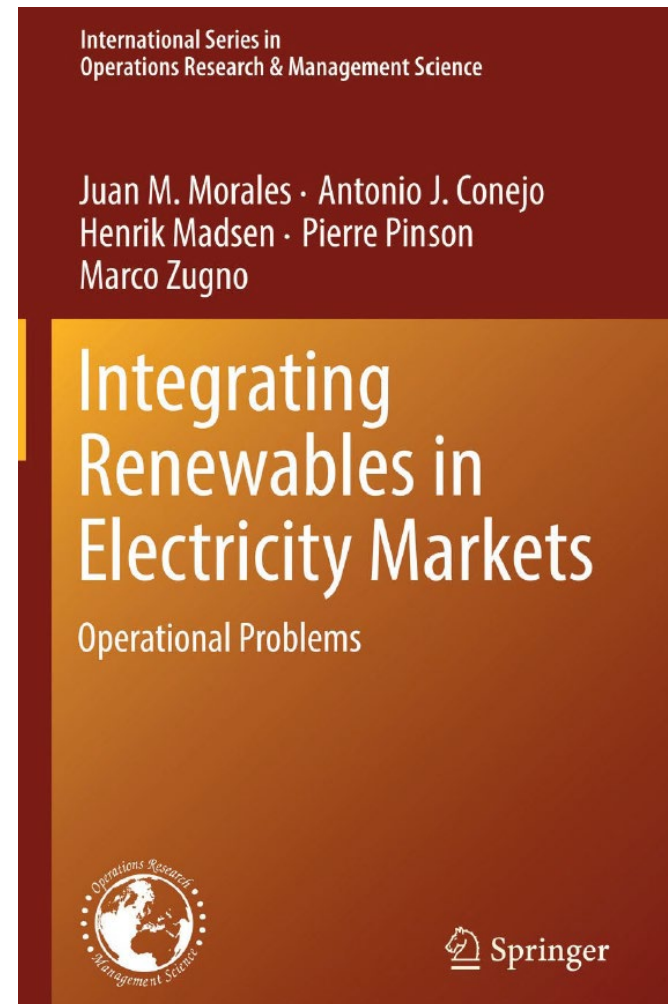
Electricity Market

Goal: meeting demand at the maximum social welfare (or the minimum total operational cost)

- **Renewables** (with stochastic generation) bring uncertainty: inaccurate forecast may result in wrong commitment and dispatch decisions, with an **increased** total operational cost for the system
- **How to manage renewable power uncertainty:**
 - ✓ Flexibility integration (fast generators, storage, demand response, etc)
 - ✓ Proper market design
- **In hydro dominated system dealing with uncertainty is an important part of hydro scheduling task**

Reference

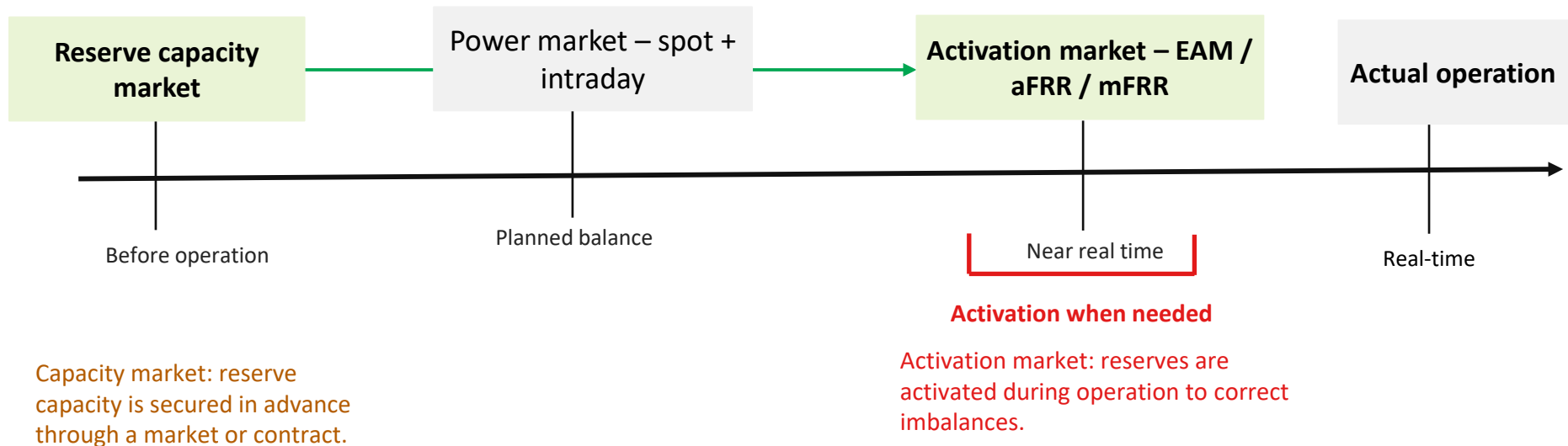
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Henrik Madsen, Pierre Pinson, Marco
Zugno, “Integrating Renewables in
Electricity Markets”, 2014, Volume
205, ISBN : 978-1-4614-9410-2



Power markets are supported by reserve markets

From planned balance to actual operation: Statnett secures capacity in advance and activates reserves when imbalances occur.

- ❑ The spot and intraday markets plan the system balance before the operating quarter/hour.
- ❑ Deviations in generation or consumption, as well as system faults, create imbalances that must be managed in real-time.
- ❑ Statnett procures reserves in capacity markets and activates them automatically or through activation markets when needed.



Clearing Energy and Reserve Capacity Sequentially

(Reserve capacity first, then day-ahead energy)

G_A	G_B	Load and reserve
• Capacity: 100 MW • Production cost: 10 €/MWh • Reserve cap. Cost: 0 €/MW	• Capacity: 50 MW • Production cost: 30 €/MWh • Reserve cap. Cost: 5 €/MW	• Load: 120 MW • Rationing cost: 200 €/MWh • Reserve requirement: 20 MW

- The system operator estimates that 20 MW of reserve capacity are required.
- let us suppose that energy and reserve capacity are *sequentially* settled in this order.
 - **First**, the reserve capacity market is cleared
 - **Next**, the energy-only dispatch is determined

1- Reserve capacity market



2- Day-ahead energy market

R_A and R_B : reserve capacity sold by G_A and G_B

P_A and P_B : energy sold by G_A and G_B

Key point: the reserve-first market sees reserve cost only, not the opportunity cost of using cheap energy capacity for reserves.

Sequential Markets: Reserve Capacity → Energy

Reserve capacity market

$$\text{Minimise}_{R_A, R_B} (0R_A + 5R_B)$$

s.t.

$$R_A + R_B = 20, (\lambda_R)$$

$$0 \leq R_A \leq 100$$

$$0 \leq R_B \leq 50$$

$$R_A^* = 20 \text{ MWh}$$

$$R_B^* = 0 \text{ MWh}$$

$$\lambda_{DA}^* = 0 \text{ €/MWh}$$

The reserve market chooses G_A because its reserve offer is cheapest.

Day-ahead energy market

$$\text{Minimise}_{P_A, P_B, P_{rat}} (10P_A + 30P_B + 200P_{rat})$$

s.t.

$$P_A + P_B + P_{rat} = 120, (\lambda_{DA})$$

$$0 \leq P_A \leq 100 - R_A^* = 80$$

$$0 \leq P_B \leq 50 - R_B^* = 50$$

$$0 \leq P_{rat} \leq 120$$

$$P_A^* = 80 \text{ MWh}$$

$$P_B^* = 40 \text{ MWh}$$

$$P_{rat} = 0 \text{ MW}$$

$$\lambda_{DA}^* = 30 \text{ €/MWh}$$

But this leaves only 80 MW of G_A available for energy, so G_B must generate 40 MWh.

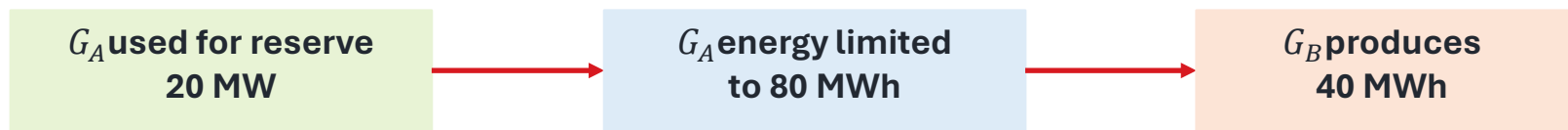
Sequential Markets: Total Cost

Total system operation cost

$$\begin{aligned} TC^{\text{seq}} &= 10P_A^* + 30P_B^* + 0R_A^* + 5R_B^* \\ &= 10(80) + 30(40) + 0(20) + 5(0) = 2100\text{€} \end{aligned}$$

Why is this inefficient?

The reserve-first market buys reserve from G_A because it is free. However, G_A is also the cheapest energy generator. Reserving 20 MW from G_A forces the day-ahead market to dispatch more expensive G_B energy.



Co-optimisation of Energy and Reserve capacity

Energy and reserve capacity are cleared simultaneously in one auction.

$$\underset{P_A, P_B, R_A, R_B, P_{rat}}{\text{Minimise}} (10P_A + 30P_B + 0R_A + 5R_B + 200P_{rat})$$

s.t.

$$P_A + P_B + P_{rat} = 120, (\lambda_{DA})$$

$$R_A + R_B \geq 20, (\lambda_R)$$

$$P_A + R_A \leq 100, P_A \geq 0, R_A \geq 0$$

$$P_B + R_B \leq 50, P_B \geq 0, R_B \geq 0$$

$$P_{rat} \leq 120, P_{rat} \geq 0$$

$$P_A^* = 100 \text{ MWh}, R_A^* = 0 \text{ MWh}$$

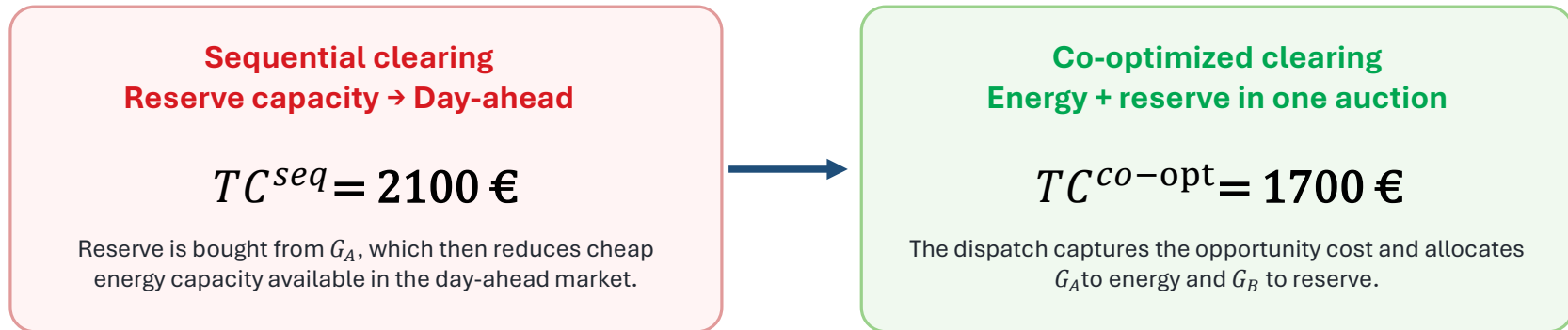
$$P_B^* = 20 \text{ MWh}, R_B^* = 20 \text{ MWh}$$

$$P_{rat}^* = 0 \text{ MWh}$$

$$\lambda_{DA}^* = 30 \text{ €/MWh}, \lambda_R^* = 5 \text{ €/MWh}$$

$$\text{TC}^{co-opt} = 10P_A^* + 30P_B^* + 0R_A^* + 5R_B^* = 10(100) + 30(20) + 0(20) + 5(20) = 1700 \text{ €}$$

Benefit of Energy–Reserve Co-optimization



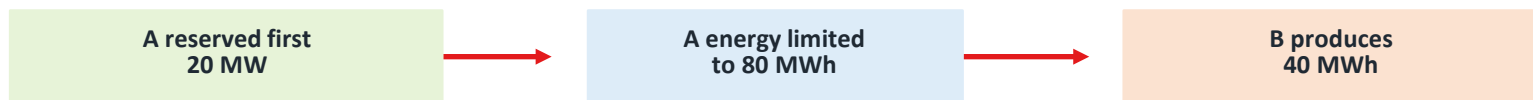
Cost reduction

$$\Delta TC = TC^{seq} - TC^{co-opt} = 400 \text{ €}$$

Co-optimization lowers total cost because it accounts for the coupling between energy and reserve capacity and avoids reserving cheap energy capacity when it is more valuable for energy dispatch.

Notes on Co-optimisation of Energy and Reserve Capacity

- **The interaction between energy and reserve capacity is seen by comparing sequential and co-optimised clearing:**
 - Sequential clearing first buys 20 MW of reserve from Producer A, because A has the cheapest reserve offer. This leaves only 80 MWh of A's cheap energy capacity for the day-ahead market.
 - Producer B, which has more expensive energy, must then produce 40 MWh to meet the 120 MWh demand. The total sequential cost is therefore 2000 €.
 - In the co-optimised clearing, Producer A is used for 100 MWh of cheap energy, while Producer B provides 20 MWh of energy and 20 MW of reserve. The total co-optimised cost is 1700 €, giving a 300 € saving.
 - The key mechanism is the opportunity cost of reserving A's capacity: using 1 MW of A for reserve can force replacement by B in energy, costing $30 - 10 = 20$ €/MW.
 - The co-optimised reserve price is 5 €/MW in this example, because B's reserve offer is marginal once A's energy value is accounted for.



Case study with wind power

- We assume that G_A can only deliver power to the day-ahead market
- The system operator commissioned wind power with the installed capacity of 80 MW

G_A 	G_B 	WP 	Load and reserve 
• Capacity: 100 MW • Production cost: 10 €/MWh • Inflexible generator	• Capacity: 40 MW • Production cost: 30 €/MWh • Flexible: reserve cap. Cost: 25 €/MW	• Capacity 80 MW • Production cost: 0 €/MWh	• 120 MW (inelastic and deterministic) • Rationing cost: 200 €/MWh • Reserve requirement: 20 MW

We assume that according to the best forecast for wind the realisation in a real-time production is **30** MW

Co-optimisation of Energy and Reserve Capacity (Prefect forecast for wind)

G_A	G_B	WP	Load and reserve
- Capacity: 100 MW - Production cost: 10 €/MWh - Inflexible generator	- Capacity: 40 MW - Production cost: 30 €/MWh - Flexible: reserve cap. Cost: 25 €/MW	- Capacity 80 MW - Production cost: 0 €/MWh	120 MW (inelastic and deterministic) - Rationing cost: 200 €/MWh - Reserve requirement: 20 MW

Wind Forecast in DA [MW]	DA+R market outcome	Balancing Market (BM)	Total operational cost (€)
30 (perfect forecast)	WP=30 MW G_A = 90 MW G_B =0 MW R_B =20 MW Cost= 1400 €	Δ WP=unchanged Δ G_A = unchanged Δ G_B =unchanged Cost= 0 €	CostDA+R+CostBM =1400 €

Despite all advances in forecasting methods, the wind forecast can still be not perfect!

Let's now consider an *imperfect* wind power forecast in the DA stage!

Co-optimisation of Energy and Reserve capacity (Imprefect forecast for wind)

G_A	G_B	WP	Load and reserve
- Capacity: 100 MW - Production cost: 10 €/MWh - Inflexible generator	- Capacity: 40 MW - Production cost: 30 €/MWh - Flexible: reserve cap. Cost: 25 €/MW	- Capacity 80 MW - Production cost: 0 €/MWh	120 MW (inelastic and deterministic) - Rationing cost: 200 €/MWh - Reserve requirement: 20 MW

Wind Forecast in DA [MW]	DA+R market outcome	Balancing Market (BM)	Total operational cost
40 deficit power of 10 MW in RT	WP=40 MW G_A = 80 MW G_B =0 MW R_B =20 MW Cost= 1300 €	ΔWP = -10 MW (production deficit) ΔG_A = unchanged ΔG_B = +10 MW (up regulation) Cost= 300 €	$\text{Cost}^{\text{DA+R}} + \text{Cost}^{\text{BM}} = 1600 \text{ €}$

Real-time wind production will be 30 MW!

Co-optimisation of Energy and Reserve capacity (Imprefect forecast for wind)

G_A	G_B	WP	Load and reserve
- Capacity: 100 MW - Production cost: 10 €/MWh - Inflexible generator	- Capacity: 40 MW - Production cost: 30 €/MWh - Flexible: reserve cap. Cost: 25 €/MW	- Capacity 80 MW - Production cost: 0 €/MWh	120 MW (inelastic and deterministic) - Rationing cost: 200 €/MWh - Reserve requirement: 20 MW

Wind Forecast in DA [MW]	DA+R market outcome	Balancing Market (BM)	Total operational cost
80 deficit power of 50 MW in RT	WP=80 MW $G_A = 40$ MW $G_B = 0$ MW $R_B = 20$ MW Cost= 900 €	$\Delta WP = -50$ MW (production deficit) $\Delta G_A =$ unchanged $\Delta G_B = +40$ MW (up regulation) Load curtailed: +10 MW Cost= 3200 €	$\text{Cost}^{\text{DA+R}} + \text{Cost}^{\text{BM}} = 4100$ €

Real-time wind production will be 30 MW!

Co-optimisation of Energy and Reserve capacity (Imprefect forecast for wind)

G_A	G_B	WP	Load and reserve
- Capacity: 100 MW - Production cost: 10 €/MWh - Inflexible generator	- Capacity: 40 MW - Production cost: 30 €/MWh - Flexible: reserve cap. Cost: 25 €/MW	- Capacity 80 MW - Production cost: 0 €/MWh	120 MW (inelastic and deterministic) - Rationing cost: 200 €/MWh - Reserve requirement: 20 MW

Wind Forecast in DA [MW]	DA+R market outcome	Balancing Market (BM)	Total operational cost (€)
5 excess power of 25 MW in RT	WP=5 MW G_A = 100 MW G_B =15 MW R_B =20 MW Cost= 1950 €	Δ WP=25MW(production excess) Δ G_A = unchanged Δ G_B =-15 MW(down regulation) Wind spilled: +10 MW Cost= -450 €	$\text{Cost}^{\text{DA+R}} + \text{Cost}^{\text{BM}} = 1550$

Real-time wind production will be 30 MW!

Summary of results

Forecast	Wind forecast in DA+R [MW]	DA+R market [€]	Balancing market[€]	Total operational cost [€]
perfect	30	1400	0	1400
Imperfect	40	1300	300	1600
	80	900	3200	4100
	5	1950	-450	1550

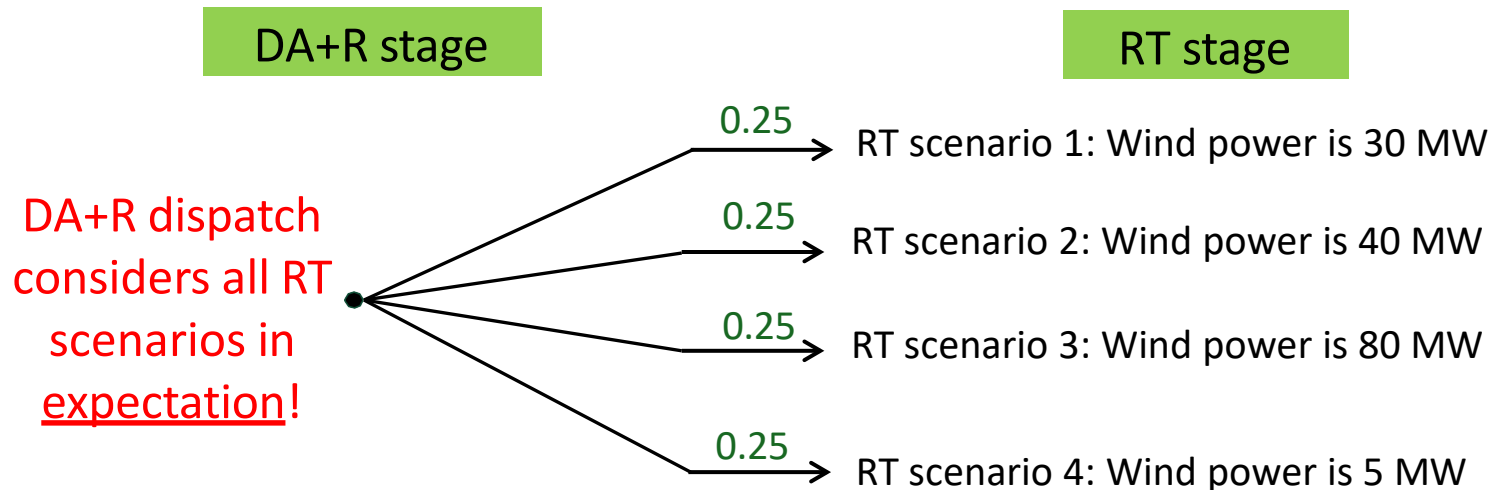
Real-time wind production will be 30 MW!

- Which wind forecast should be considered in DA+R?
- When we clear day-ahead market, we do not know the actual realisation of Wind
- We portray the future as possible scenarios

Discussion

- In DA+R stage:

Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



Stochastic Programming Problems with Recourse



Stages

Problems are classified by their number of stages.



A stage = a decision point

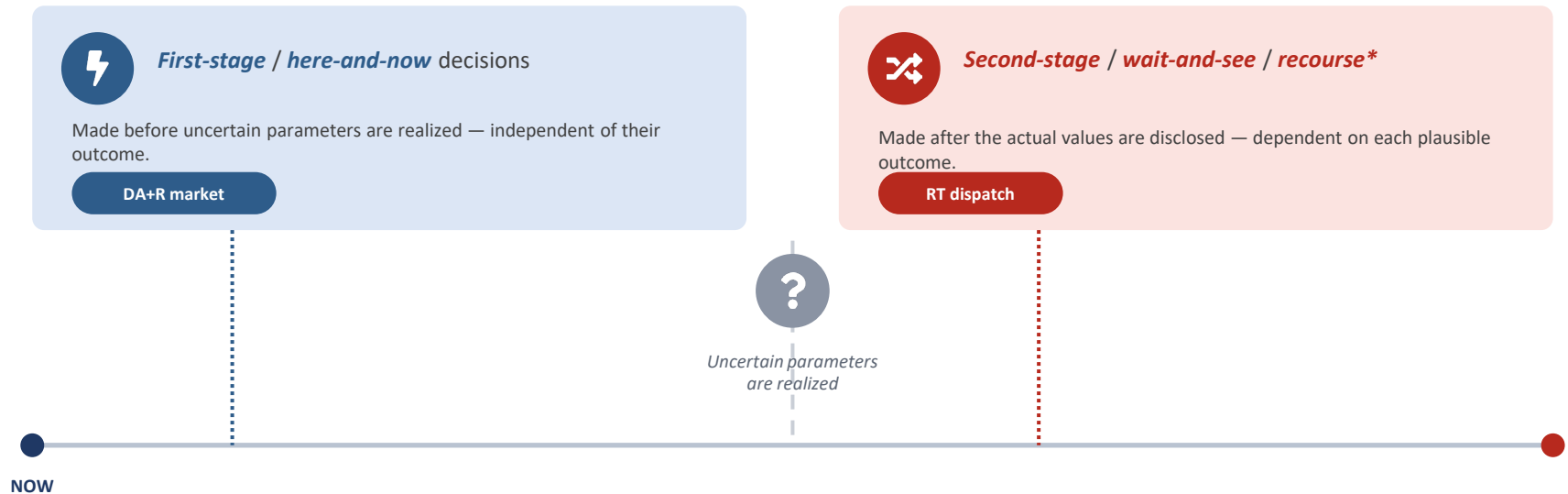
The point in time when new information is received.



of stages

Depends on how decisions & information disclosure are sequenced.

The Simplest Case: Two-Stage Stochastic Programming



* **Recourse** points to the fact that second-stage decisions enable the decision-maker to adapt to the actual outcomes of the random events.

Two-stage stochastic Problem

Scenario tree for a two-stage problem



First-stage / here-and-now decisions



Second-stage / wait-and-see decisions

Reading a scenario tree



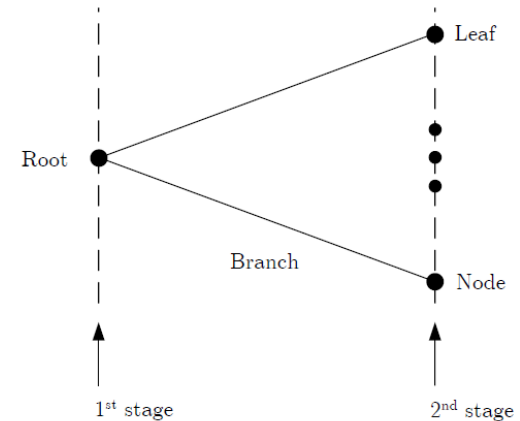
Root Starting point — the state known today



Branch A link representing one possible transition

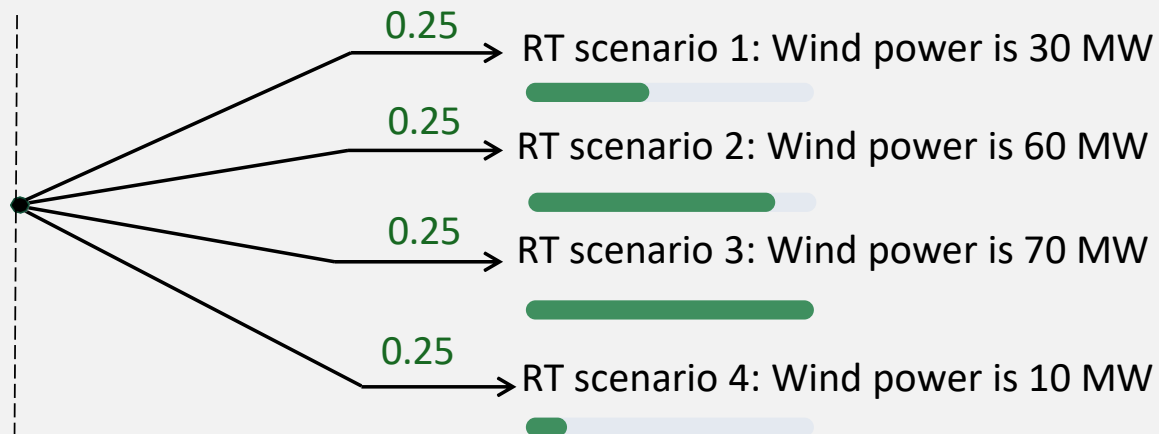


Node/Leaf A point where the tree splits into outcomes



DA+R stage (first)

RT stage (second)



Stochastic market-clearing mechanism

● In DA stage:

Look-ahead strategy



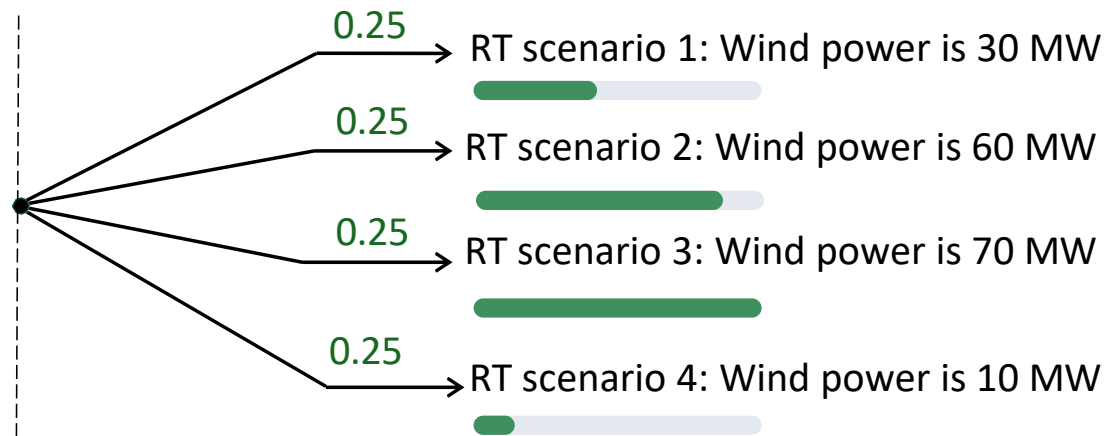
Expectation-based dispatch

DA+R dispatch considers **all RT scenarios** at once, in expectation!

$$\mathbb{E}[\text{Cost}] = \sum_{\omega \in \Omega} \pi_{\omega} \text{Cost}(\omega)$$

DA+R stage

RT stage



The day-ahead dispatch is never optimized against a single guess of wind output, it is hedged against *every* scenario the tree contains, weighted by how likely each one is.

Mathematical Formulation: Two-Stage Stochastic Problem

$$\underset{x, y_\omega}{\text{Minimise}} \left(\underbrace{c^T x}_{\substack{\text{First-stage cost} \\ \text{(paid regardless of } \omega)}} + \sum_{\omega \in \Omega} \pi_\omega \underbrace{q_\omega^T y_\omega}_{\substack{\text{Expected recourse cost} \\ \text{(probability-weighted over all } \omega \in \Omega)}} \right) \quad (1.1)$$

subject to

$$Ax \leq b \quad (1.2) \quad \text{FIRST-STAGE} \quad \text{Feasibility of today's decision } x, \text{ independent of any scenario.}$$

$$W_\omega y_\omega \leq h_\omega - T_\omega x, \quad \forall \omega \in \Omega \quad (1.3) \quad \text{RECOURSE LINK} \quad \text{The key recourse constraint: } y_\omega \text{ must react so that, together with } x, \text{ feasibility holds in every scenario } \omega.$$

$$x \geq 0, \quad (1.4) \quad \text{FIRST-STAGE} \quad \text{First-stage variables are non-negative.}$$

$$y_\omega \geq 0, \quad \forall \omega \in \Omega \quad (1.5) \quad \text{SECOND-STAGE} \quad \text{Second-stage variables are non-negative, in every scenario.}$$

x : set of **first-stage** variables (**scenario-independent**)

π_ω : probability of occurrence of scenario ω

y_ω : set of **second-stage** variables (**scenario-dependent**)

$c, q_\omega, b, h_\omega, A, T_\omega, W_\omega$ are known vectors and matrices of appropriate size



This formulation is the **deterministic equivalent problem** of the stochastic program: all randomness is folded into extra variables and constraints, so it can be solved directly by off-the-shelf LP software.

A Generalization of Stochastic Market-Clearing

Minimize

⚡ total operational cost¹ of the system in **DA+R**

+

⚡ total **expected** operational cost of the system balancing in **RT**

subject to:

⚡ Day-Ahead & Reserve Market Constraints

- Power balance equations at the day-ahead stage
- Reserve capacity determination constraints
- Bounds of reserve and energy offers

⚡ Real-Time Operation Constraints

- Power balance equations at the balancing stage
- Deployed reserve determination constraints

⚡ *Non-negativity declarations for all decision variables*

¹ We assume loads are inelastic to price; otherwise the social welfare should be maximized.

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

DA+R variables: schedules of G_A , G_B and WP

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

RT variables: flexible unit adjustment under scenarios s1 to s4

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

RT variables: wind power spillage under scenarios s1 to s4

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{SP} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

RT variables: load shedding under scenarios s1 to s4

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

Total operational cost of the system in DA+R

Total expected operational cost of the system in RT

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \underbrace{\sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d)}_{\text{Total operational cost of the system in DA+R}} + \underbrace{\sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)}_{\text{Total expected operational cost of the system in RT}}$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

DA constraints

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

RT constraints under scenarios s1 to s4

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

Wind power realization under scenarios s1 to s4

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (\boxed{w_s^{RT}} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq \boxed{w_s^{RT}}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

Constraints linking DA and RT variables

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA}) \quad \text{DA+R market price}$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

Probability-weighted RT market price under scenarios s1 to s4

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Stochastic market-clearing

	DA+R	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 40 MW	RT (scenario 3) Wind power realization: 80 MW	RT (scenario 4) Wind power realization: 5 MW
G_A [MW]	100	unchanged	unchanged	unchanged	unchanged
G_B [MW]	15	-15 (down regulation)	-15 (down regulation)	-15 (down regulation)	unchanged
WP [MW]	5	+15 (10 MW spilled)	+15 (20 MW spilled)	+15 (60 MW spilled)	unchanged
R_B^u [MW]	0	-	-	-	-
R_B^d [MW]	15	-15	-15	-15	-
Load [MW]	120	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)
Corresponding operational cost [€]	1600	-450	-450	-450	0

Total expected operational cost of the system

$$= 1600 + 0.25 * [-450 - 450 - 450 + 0] = 1262.5 \text{ €}$$

Why Out-of-Sample Analysis Matters

 A decision can look optimal on the scenarios it was built from **and still perform poorly** once reality plays out differently.



950 in-sample scenarios

Used to optimize the DA decision

*fixed DA
decision*



50 out-sample scenarios

Used to judge that decision, honestly



Exposes overfitting

The solution may fit the 950 scenarios too closely instead of capturing the true uncertainty..



Gives an unbiased estimate

Evaluating on scenarios the model never optimized against yields a realistic expected cost, not a flattering one.



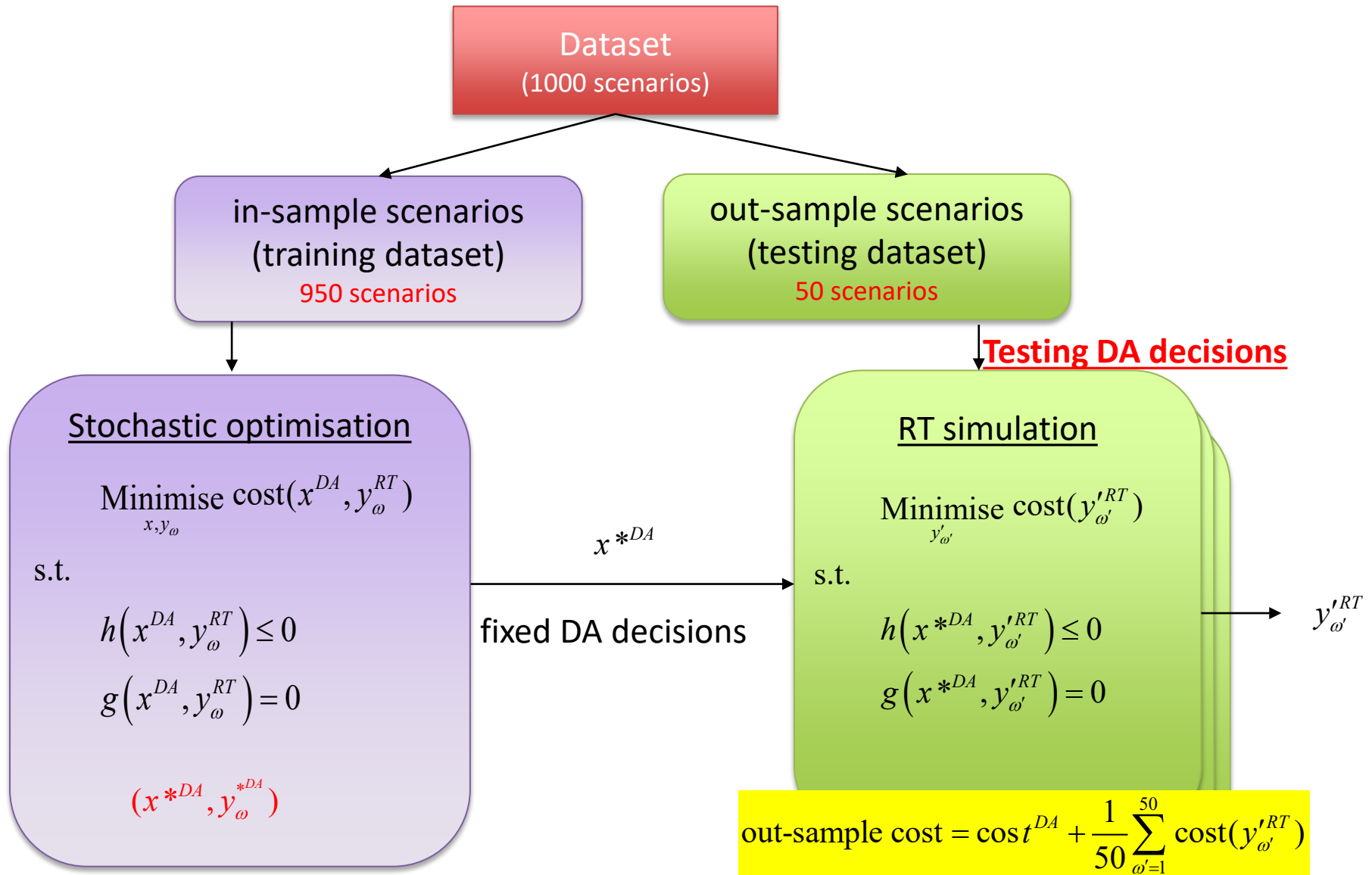
Builds trust before deployment

Operators need evidence the DA decision holds up on outcomes it hasn't seen — not just the ones it was trained on.



Rule of thumb: never grade a decision using the scenarios that produced it — always test it on scenarios it has never seen.

An out-of-sample analysis



Mathematical formulation: Stochastic market-clearing with fixed reserve requirement

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} R_k^u \geq R \quad \text{fixed reserve requirement}$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Stochastic market-clearing

	DA+R	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 40 MW	RT (scenario 3) Wind power realization: 80 MW	RT (scenario 4) Wind power realization: 5 MW
G_A [MW]	95	unchanged	unchanged	unchanged	unchanged
G_B [MW]	0	unchanged	unchanged	unchanged	+20 (up regulation)
WP [MW]	25	0 (5 MW spilled)	0 (15 MW spilled)	0 (55 MW spilled)	+20 (0 MW spilled)
R_B^u [MW]	20	-	-	-	+20
R_B^d [MW]	0	-	-	-	-
Load [MW]	120	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)
Corresponding operational cost [€]	1450	0	0	0	600

Total expected operational cost of the system
 $= 1600 + 0.25 \cdot [0 + 0 + 0 + 600] = 1750 \text{ €}$